

Fine-Tuning Quality Over Model Size: SLM-to-SLM Guided Reasoning with Statistically Verified Gains

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Abstract

We investigate whether a fine-tuned small language model (SLM) can replace expensive large-model guides in a reasoning pipeline. Our pipeline, GUIDEDVOTE, pairs a LoRA fine-tuned 3B *guide* that produces a structured verbal reasoning plan with a 1B–1.5B *solver* that casts five independent votes before majority-vote decoding. The combined compute cost is 10.5B parameter-passes, 30–47% below the natural 15B ceiling. Evaluated across seven benchmarks spanning arithmetic, commonsense, science, and reading comprehension, GUIDEDVOTE yields statistically confirmed accuracy gains on six datasets using McNemar’s test ($\alpha = 0.05$), with improvements of up to +24.2 percentage points. An ablation rules out model size (untuned 3B guide is neutral or harmful) and chain-of-thought prompting (equal-compute CoT fails on four of five datasets) as sufficient explanations, leaving fine-tuning quality as the necessary condition. A cross-architecture replication with Llama 3.2 (3B guide, 1B solver) confirms that arithmetic reasoning gains are architecture-robust (+21.0 pp on SVAMP), while a large physical-reasoning gain observed on PIQA with one solver is identified as a position-bias correction artefact rather than a general reasoning improvement. A clean null on reading-comprehension (RACE-High, $p = 1.000$ with the primary model pair) defines the method’s boundary: verbal planning helps tasks requiring solution generation, not evidence retrieval.

1 Introduction

The dominant strategy for improving small language model (SLM) reasoning is to scale model size or distill from a much larger teacher (Ho et al., 2023; Magister et al., 2023). Both strategies incur substantial compute or data costs. We ask a different question: *can a fine-tuned SLM of moderate size guide a smaller SLM, replacing the expensive LLM guide without sacrificing performance?*

We propose GUIDEDVOTE, a two-stage pipeline in which (i) a 3B guide generates a structured verbal reasoning plan from the question, and (ii) a 1.5B (or 1B) solver reads the plan and produces five independent answers via temperature sampling, with the majority vote as the final prediction. The guide is fine-tuned with LoRA (Hu et al., 2022) on GSM8K (Cobbe et al., 2021); the solver is frozen. The total cost is $3B + 5 \times 1.5B = 10.5B$ parameter-passes, 30% below the 15B ceiling of running the 3B model alone for all five votes.

Our contributions are:

- **Empirical breadth.** We evaluate across seven benchmarks (six domains) with 2,300+ question evaluations and apply McNemar’s paired significance test with power analysis for each dataset.
- **Three-pronged elimination argument.** Ablations show that neither scaling the guide (untuned 3B) nor prompting it (chain-of-thought at equal compute) replicates the gains of fine-tuned verbal planning.
- **Mechanistic findings.** We identify position-bias suppression as the key mechanism behind the largest single gain and show that retrieval tasks form a principled boundary condition.
- **Cross-architecture replication.** A full replication with Llama 3.2 models confirms architecture-robust and architecture-conditional gains, tightening the scope of each claim.

2 Related Work

Chain-of-thought prompting. Wei et al. (2022) showed that asking LLMs to reason step-by-step substantially improves multi-step reasoning. Subsequent work extended this to least-to-most prompting (Zhou et al., 2023), self-consistency decoding (Wang et al., 2023), and tree-of-thought search

(Yao et al., 2023). Our work shows that for SLMs, unstructured “think step by step” prompting is insufficient or harmful on four of five datasets, motivating the use of a fine-tuned plan-generator instead.

Self-consistency and ensemble decoding. Wang et al. (2023) established that majority voting over multiple sampled reasoning paths improves accuracy without requiring additional training. We adopt this ensemble as our solver’s decoding strategy and show that a fine-tuned guide systematically shifts the vote distribution toward correct answers, increasing what we term *vote consistency*.

Knowledge distillation for small models. Ho et al. (2023) and Magister et al. (2023) distill reasoning chains from large models into smaller ones. Our approach differs: the solver is never fine-tuned; only the guide is, and on a single general dataset (GSM8K). This makes deployment cheaper and the guide reusable across solvers.

Model collaboration pipelines. Li et al. (2023) and Du et al. (2024) explore multi-agent reasoning with LLMs. Fu et al. (2023) decompose reasoning into specialised roles. Our work brings this paradigm to the SLM regime and provides rigorous statistical evaluation absent from most prior collaborative-agent work.

Calibration of language models. Desai and Durrett (2020) and Kadavath et al. (2022) study whether model confidence reflects accuracy. We measure Expected Calibration Error (ECE) (Naeini et al., 2015) before and after guidance and find universal ECE improvement across all seven datasets, including those where accuracy does not change.

3 Method

3.1 Pipeline Overview

Given a question q , GUIDEDVOTE operates in two stages:

Stage 1 — Plan generation. The fine-tuned guide model $\mathcal{G}$ receives q and produces a structured verbal plan ρ :

$$\rho = \mathcal{G}(q), \quad \mathcal{G} = \mathcal{M}_{3B} + \text{LoRA}.$$

The plan uses an elimination format: it lists candidate options, rules out distractors with explicit reasoning steps, and proposes a preferred answer. Temperature is set to 0.1 for deterministic planning.

Stage 2 — Guided voting. The frozen solver $\mathcal{S}$ receives the concatenation (q, ρ) and is sampled $K = 5$ times at temperature 0.4. The final answer is the majority vote:

$$\hat{y} = \text{MajorityVote} \left\{ \mathcal{S}(q, \rho)^{(k)} \right\}_{k=1}^5.$$

Ties (fewer than 3 votes for any single option) are broken by a single deterministic refiner call ($T = 0.1$).

Baseline. The solver $\mathcal{S}$ is sampled $K = 5$ times on q alone (no plan, same majority-vote decoding).

3.2 Compute Regime

Table 1 shows the parameter-pass budget. The guided pipeline sits between the baseline and the natural ceiling (running the 3B model for all five votes).

Setup	Param-passes	vs. ceiling
Baseline (1.5B $\times$ 5)	7.5B	−50%
GUIDEDVOTE (3B + 1.5B $\times$ 5)	10.5B	−30%
Ceiling (3B $\times$ 5)	15.0B	—
Baseline (1B $\times$ 5) [Llama]	5.0B	−67%
GUIDEDVOTE (3B + 1B $\times$ 5) [Llama]	8.0B	−47%

Table 1: Compute budget for both model pairs. GUIDEDVOTE operates at 30–47% below the natural ceiling depending on solver size.

3.3 Fine-Tuning the Guide

We fine-tune Qwen2.5-3B (and Llama 3.2-3B in the replication) using LoRA (rank 16, $\alpha = 32$, dropout 0.05) on the GSM8K training set (Cobbe et al., 2021) for three epochs. Training examples are reformatted to elicit the elimination-style plan format: the model is trained to enumerate options, reason about each, eliminate distractors, and conclude with a recommendation. No task-specific fine-tuning is performed on any evaluation benchmark.

4 Experimental Setup

Models. Primary pair: Qwen2.5-3B (guide, LoRA fine-tuned) and Qwen2.5-1.5B (solver, frozen). **Replication pair:** Llama 3.2-3B (guide, LoRA fine-tuned) and Llama 3.2-1B (solver, frozen). Both guides use the same GSM8K fine-tuning procedure.

Benchmarks. We evaluate on seven benchmarks across six domains: **SVAMP** (Patel et al., 2021) and **ASDiv** (Miao et al., 2020) (arithmetic math);

PIQA (Bisk et al., 2020) (physical reasoning); **CommonsenseQA** (Talmor et al., 2019) (commonsense); **ARC-Challenge** (Clark et al., 2018) (science reasoning); **AQUA-RAT** (Ling et al., 2017) (algebraic MCQ); and **RACE-High** (Lai et al., 2017) (reading comprehension). Sample sizes range from 100 to 900; all sampling is seeded for reproducibility.

Evaluation. We report raw accuracy and three additional angles: (i) **vote consistency** = $\frac{1}{N} \sum_q \frac{\text{correct votes}_q}{K}$, (ii) **ECE** (Naeini et al., 2015) using vote agreement as confidence proxy, and (iii) McNemar’s chi-square test (with Yates’ continuity correction) on the paired correct/incorrect vectors. All p -values are two-sided; significance threshold $\alpha = 0.05$.

5 Results

5.1 Primary Results (Qwen Pair)

Table 2 presents the main results. Six of seven datasets show accuracy gains; four are statistically confirmed ($p < 0.01$); one (RACE-High) is a clean null.

Vote consistency. The mean fraction of correct votes per question rises on every confirmed dataset, with SVAMP showing the largest lift ($1.90\times$). High-agreement questions (4–5/5 votes correct) increase substantially on PIQA ($269 \rightarrow 381$) and SVAMP ($18 \rightarrow 115$), indicating that the guide shifts the solver toward reliable rather than lucky majorities.

Calibration. ECE improves on all seven datasets including RACE-High where accuracy is unchanged. A sign test on the ECE direction gives $p = 2^{-7} = 0.0078$, confirming calibration improvement is not by chance. The most dramatic case is PIQA baseline: 491 of 500 questions have 100% vote agreement (all choosing option A), yet accuracy is only 54.2% — maximum confidence, near-random accuracy. After guidance, ECE drops from 0.354 to 0.110 (−69%).

5.2 Power Analysis

For each dataset we compute the minimum N required for 80% power using the McNemar power formula (Lachenbruch, 1981):

$$N_{\min} = \left(\frac{z_{\alpha/2} \sqrt{\pi_d} + z_{\beta} \sqrt{\pi_d - \delta^2}}{\delta} \right)^2,$$

where $\pi_d = \pi_B + \pi_C$ is the discordant rate and $\delta = \pi_B - \pi_C$. All four confirmed datasets exceeded their required N . RACE-High ($N = 500$, power $\approx 100\%$) provides strong evidence for a true null: any real effect ≥ 5 pp would have been detected.

6 Ablation: Fine-Tuning vs. Size vs. Prompting

6.1 Model-Size Ablation

We replace the fine-tuned 3B guide with the *same model before fine-tuning* (base 3B, no LoRA), keeping all other conditions identical. Table 3 shows results on four datasets.

The untuned guide creates a “confident misdirection” failure mode on CommonsenseQA: false-confidence cases (5/5 votes on the wrong answer) rise from 54 to 65 relative to baseline. The structured elimination format from fine-tuning prevents this.

6.2 Chain-of-Thought Baseline Control

We run a third condition: the solver receives a “think step by step” suffix instead of the fine-tuned plan, at identical compute (7.5B parameter-passes, no guide cost). Table 4 shows the three-way comparison.

This yields a three-tier task taxonomy: **Tier 1** (RACE-High): neither CoT nor guidance helps; the answer requires evidence retrieval, not solution generation. **Tier 2** (ARC-Challenge): CoT is sufficient; domain knowledge is directly accessible through prompting (CoT = guided = 80.0%, ECE = 0.099). **Tier 3** (PIQA, CSQA, SVAMP, ASDiv): CoT is insufficient or harmful; fine-tuned verbal planning provides +8–25 pp beyond CoT.

PIQA position bias. The CoT condition breaks the 1.5B Qwen solver’s A-option collapse (92.4% $\rightarrow$ 44.2% sol-1 selection) yet accuracy barely changes (55.4% $\rightarrow$ 54.8%). GUIDEDVOTE achieves 78.4% at a similar sol-1 rate (42.1%). Debiasing is therefore necessary but not sufficient; fine-tuned content-grounded reasoning provides the remaining +23.6 pp.

7 Analysis

7.1 Position Bias Suppression

We define position bias as systematic over-selection of a single option across questions regardless of content. Two datasets exhibit measurable bias:

Dataset	Domain	N	Base	Guided	Δ	Consist. lift	ECE imp.	McNemar p
PIQA	Physical	500	54.2	78.4	+24.2	1.43×	69.1%	≈ 0 [***]
SVAMP	Arithmetic	300	40.3	61.7	+21.3	1.90×	36.6%	≈ 0 [***]
ARC-Challenge	Science	900	71.7	80.0	+8.3	1.09×	32.6%	< 0.001 [**]
ASDiv	Arithmetic	300	53.7	64.0	+10.3	1.18×	45.2%	0.0035 [*]
CommonsenseQA	Commonsense	900	70.7	75.8	+5.1	1.09×	19.8%	< 0.001 [**]
AQUA-RAT	Algebraic	100	40.0	52.0	+12.0	1.00×	45.6%	pending
RACE-High	Retrieval	500	58.6	58.4	−0.2	—	23.1%‡	1.000 [null]
Avg. (4 confirmed)	—	—	54.7	70.0	+15.2	1.40×	42.7%	—

Table 2: Primary results (Qwen2.5-3B guide + Qwen2.5-1.5B solver). Significance: [***] $p < 0.0001$, [**] $p < 0.001$, [*] $p < 0.01$. ECE improves on all datasets; ‡ECE improvement on RACE-High does not imply accuracy improvement (calibration universality holds).

Dataset	Base	Base 3B	FT 3B
ASDiv	57.0	56.0 (−1.0)	64.0 (+7.0)
SVAMP	39.0	61.0 (+22.0)	61.7 (+22.7)
ARC	72.0	74.0 (+2.0)	77.0 (+5.0)
CSQA	74.8	72.8 (−2.0)	77.0 (+2.2)

Table 3: Three-way ablation: unguided baseline, base (untuned) 3B guide, fine-tuned 3B guide. Delta vs. baseline in parentheses. Fine-tuning adds 5–12 pp beyond model size on all tasks; the base guide actively misleads on ASDiv and CommonsenseQA.

Dataset	N	Base	CoT	Guided	Guided vs CoT
SVAMP	300	36.7	47.0	61.7	+14.7
PIQA	500	55.4	54.8	78.4	+23.6
CSQA	900	70.4	67.7	75.8	+8.1
ARC	900	71.7	80.0	80.0	+0.0
ASDiv	300	47.6	45.8	64.0	+18.2

Table 4: Three-way comparison: baseline, CoT at equal compute (7.5B), GUIDEDVOTE at 10.5B. CoT is sufficient only for ARC-Challenge (science MCQ) where pretraining knowledge is directly accessible via prompting. On PIQA and CSQA, CoT is harmful relative to baseline.

PIQA: The unguided Qwen 1.5B solver selects sol-1 on 93.4% of questions (expected: 50%). After guidance: 42.1%, nearly uniform. Suppression: −51.3 pp, the largest observed in our study.

CommonsenseQA: Option E is selected on 24.8% of questions (expected: 20%). After guidance: 14.0% (−10.8 pp).

7.2 The RACE-High Null

$B = 60$, $C = 61$, $\chi^2 = 0.00$, $p = 1.000$ at $N = 500$. With 100% power, this constitutes strong evidence for a true null rather than an underpowered positive. The method’s boundary is principled: verbal reasoning plans scaffold solution *generation* but cannot accelerate evidence *retrieval* from a pro-

vided passage.

8 Cross-Architecture Replication

To assess generalisability, we replicate the full pipeline with Llama 3.2-3B (guide) and Llama 3.2-1B (solver), fine-tuning the guide identically on GSM8K. Table 5 presents results alongside the Qwen reference.

Architecture-robust finding. Mathematical reasoning gains (SVAMP, ASDiv) replicate with <0.3 pp difference between architectures, establishing arithmetic improvement as the most generalisable claim.

PIQA inversion. The +24.2 pp Qwen gain on PIQA becomes a statistically significant *loss* (−5.4 pp, $p = 0.031$) with Llama. Investigation shows the Llama 1B solver does not exhibit the A-option collapse seen in the Qwen 1.5B solver, removing the primary mechanism of the Qwen PIQA gain. We therefore reclassify the Qwen PIQA result as a *position-bias correction artefact*: large when the solver is biased, zero or negative when it is not.

RACE-High flip. The Qwen null ($p = 1.000$) becomes a marginal positive with Llama (+5.7 pp, $p = 0.011$). We hypothesise that the weaker Llama 1B baseline retrieval capacity allows the guide’s extractive planning to provide incremental benefit, suggesting that the retrieval boundary is determined by the solver’s pre-existing retrieval floor rather than task type alone.

Revised taxonomy. Combining both replication results, we propose a four-tier, architecture-conditional taxonomy (Table 6).

Dataset	N	Base	Guided	Δ (Llama)	Δ (Qwen ref.)	ECE imp.	McNemar p
SVAMP	300	28.7	49.7	+21.0	+21.3	45.2%	≈ 0 [***]
ARC-Challenge	900	47.6	58.1	+10.6	+10.6	35.7%	< 0.0001 [***]
ASDiv	300	39.0	50.3	+11.3	+10.3	47.9%	0.0013 [*]
CommonsenseQA	900	49.0	54.4	+5.4	+5.1	3.6%	0.0045 [*]
RACE-High	700	41.0	46.7	+5.7	-0.2	11.4%	0.011 [†]
PIQA	900	61.9	56.4	-5.4	+24.2	-25.6%	0.031 [—]

Table 5: Cross-architecture replication (Llama 3.2-3B guide + Llama 3.2-1B solver). [***] $p < 0.0001$, [*] $p < 0.01$, [†] unexpected positive, [—] guided *hurts*. SVAMP and ASDiv replicate with near-identical magnitude. PIQA inverts; RACE-High flips from null to significant.

Tier	Datasets
A: Robust	SVAMP, ASDiv
B: Accessible	ARC, CommonsenseQA
C: Bias-dep.	PIQA
D: Retrieval	RACE-High

Table 6: Revised four-tier task taxonomy derived from cross-architecture replication. Tier C collapses to a position-bias audit: measure solver option distribution before deploying the guide.

9 Conclusion

We presented GUIDEDVOTE, a pipeline in which a LoRA fine-tuned 3B guide model steers a smaller frozen solver toward more consistent and better-calibrated answers via majority voting. Across seven benchmarks and two independent model pairs, we show that (i) fine-tuning quality is the necessary condition for effective guidance — model size and chain-of-thought prompting are insufficient; (ii) confirmed accuracy gains of +5–21 pp on arithmetic and reasoning tasks are statistically robust across architectures; (iii) ECE improves universally on the primary model pair; and (iv) a clean null on reading comprehension and an identified position-bias mechanism provide principled, falsifiable boundary conditions.

The cross-architecture replication reveals that the largest individual gain (PIQA, +24.2 pp) is a solver-bias correction artefact rather than a general reasoning improvement, narrowing the central claim to a more defensible and mechanistically grounded scope. A practical decision rule follows: apply GUIDEDVOTE on arithmetic and domain-accessible reasoning; audit the solver’s option distribution before deploying on binary-choice tasks; and prefer CoT when domain knowledge is directly accessible through prompting.

Limitations

Single fine-tuning source. Both guide models are fine-tuned on GSM8K, a grade-school arithmetic dataset. Cross-domain transfer to common-sense and science benchmarks is observed empirically but the mechanism may differ from in-domain transfer. A planned experiment fine-tunes the guide on ASDiv directly to quantify domain-specific transfer gains.

English-only evaluation. All seven benchmarks are English. Whether structured verbal planning generalises to morphologically rich or low-resource languages is not addressed.

Scale ceiling. The largest model evaluated is 3B parameters. We make no claims about guide models above 7B or solver models above 3B. Interactions between guide and solver quality at larger scales remain an open question.

RACE-High architecture-conditional result. The RACE-High Llama result ($p = 0.011$) contradicts the retrieval-null claim derived from the Qwen experiments. While we provide a plausible mechanism (solver retrieval floor), this requires verification with additional model pairs before the retrieval boundary claim can be stated universally.

Refiner tiebreak. In the Llama replication, a refiner tiebreak call is used on tied votes. Its accuracy (46.9% on ARC-Challenge) is above the 25% random baseline but modest; a more principled tiebreak strategy could improve overall accuracy.

Compute measurement. We measure compute in parameter-passes, a coarse proxy. Wall-clock or FLOPs measurements under real inference infrastructure (batching, quantisation, KV-cache) may yield different efficiency conclusions.

Ethics Statement

This work uses publicly available benchmarks and open-weight language models. No new data is collected or annotated. The method could in principle reduce deployment costs for reasoning pipelines, making capable SLM-based systems more accessible. We are not aware of direct dual-use risks specific to this pipeline.

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